

DAY 4 OF 75 DAYS ETHICAL H4CKING COURSE

- Introduction
- Hacking labs setup
- Network fundamentals
- How websites works
- Linux fundamental
- Computer fundamentals
- Network security
- Wireless attacks
- System pentesting
- Social engineering
- Phishing
- Introduction to mobile hacking
- Some important tools in ethical hacking
- Wireshark, burpsuite, metasploit, autopsy, snort many more
- Introduction to webhacking
- Introduction to digital forensics
- Malware analysis
- Python basics
- Python for penetration testers
- Projects

- Introduction
- Hacking labs setup
- Network fundamentals
- websites working and technology
- Linux fundamental
- COMPUTER FUNDAMENTALS

OSI MODEL
PACKETS
PORT FORWARDING
VPN
FIREWALLS ETC

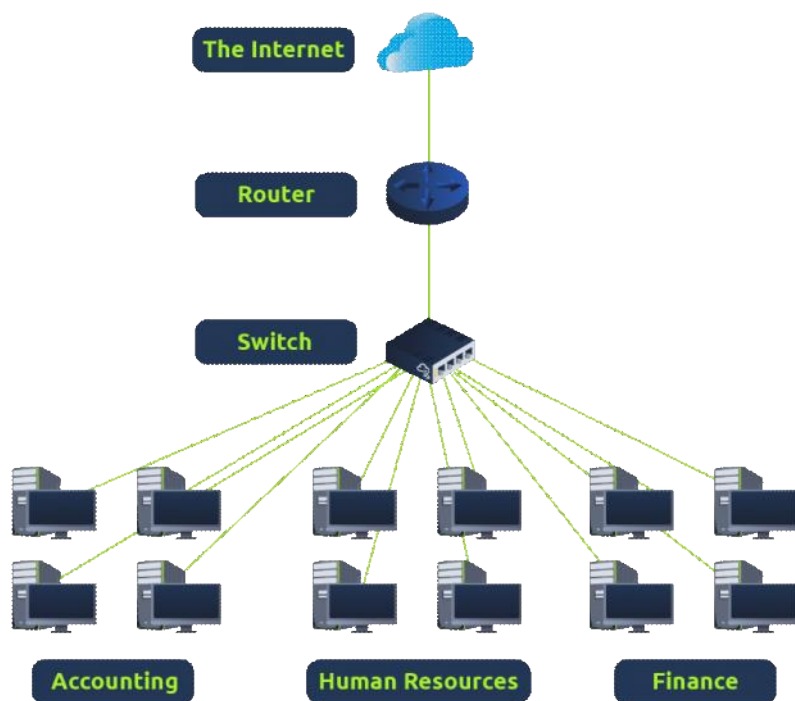
Subnetting

As we've previously discussed throughout the module so far, Networks can be found in all shapes and sizes - ranging from small to large. Subnetting is the term given to splitting up a network into smaller, miniature networks within itself.

Think of it as slicing up a cake for your friends. There's only a certain amount of cake to go around, but everybody wants a piece. Subnetting is you deciding who gets what slice & reserving such a slice of this metaphorical cake.

Take a business, for example; You will have different departments such as:

- Accounting
- Finance



Whilst you know where to send information in real life to the correct department, networks need to know as well. Network administrators use subnetting to categorise and assign specific parts of a network to reflect this.
Subnetting is achieved by splitting up the number of hosts that can fit within the network, represented by a number called a subnet mask. Let's refer back to our diagram from the first room in this module:

As we can recall, an IP address is made up of four sections called octets. The same goes for a subnet mask which is also represented as a number of four bytes (32 bits), ranging from 0 to 255 (0-255). Subnets use IP addresses in three different ways:

- Identify the network address
- Identify the host address
- Identify the default gateway

The ARP Protocol

Recalling from our previous tasks that devices can have two identifiers: A MAC address and an IP address, the Address Resolution Protocol or ARP for short, is the technology that is responsible for allowing devices to identify themselves on a network.

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Simply, the ARP protocol allows a device to associate its MAC address with an IP address on the network. Each device on a network will keep a log of the MAC addresses associated with other devices.

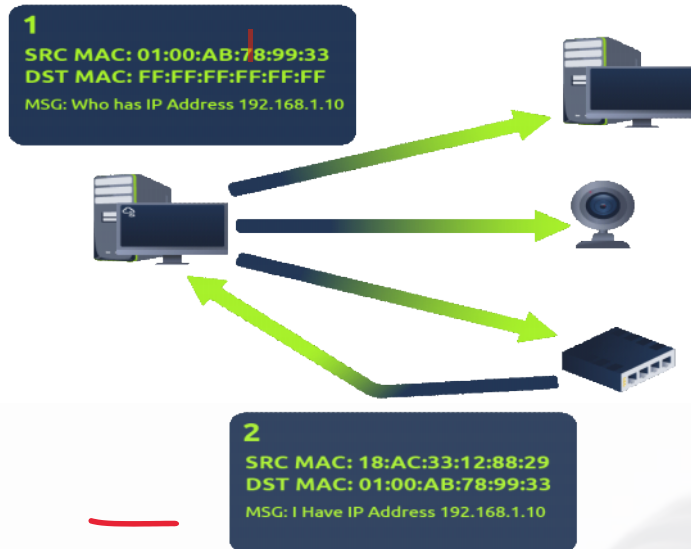
When devices wish to communicate with another, they will send a broadcast to the entire network searching for the specific device. Devices can use the ARP protocol to find the MAC address (and therefore the physical identifier) of a device for communication.

How does ARP Work?

Each device within a network has a ledger to store information on, which is called a cache. In the context of the **ARP** protocol, this cache stores the identifiers of other devices on the network. In order to map these two identifiers together (IP address and MAC address), the ARP protocol sends two types of messages:

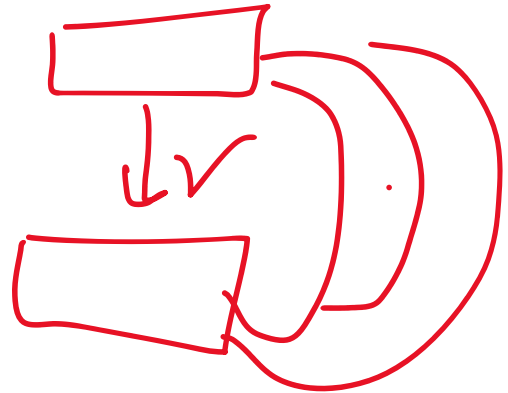
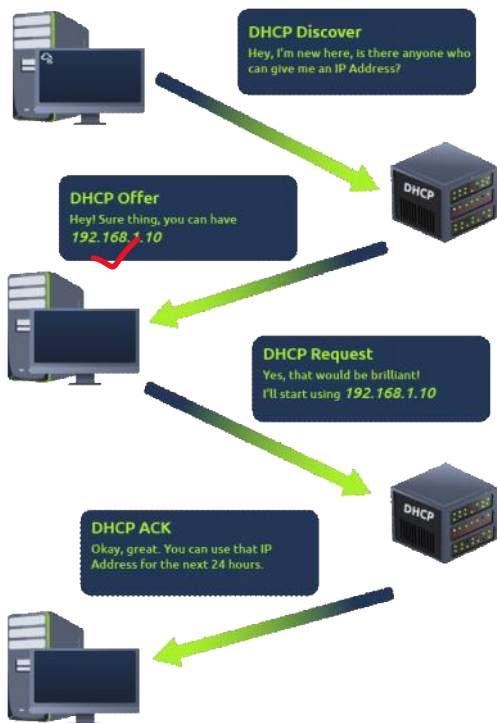
1. **ARPRequest**
2. **ARPReply**

When an **ARPRequest** is sent, a message is broadcasted to every other device found on a network by the device, asking whether or not the device's MAC address matches the requested IP address. If the device does have the requested IP address, an **ARPReply** is returned to the initial device to acknowledge this. The initial device will now remember this and store it within its cache (an **ARPEntry**). This process is illustrated in the diagram below:



The DHCP Protocol

IP addresses can be assigned either manually, by entering them physically into a device, or automatically and most commonly by using a **DHCP** (Dynamic Host Configuration Protocol) server. When a device connects to a network, if it has not already been manually assigned an IP address, it sends out a request (**DHCPDiscover**) to see if any DHCP servers are on the network. The DHCP server then replies back with an IP address the device could use (**DHCP Offer**). The device then sends a reply confirming it wants the offered IP Address (**DHCPRequest**), and then lastly, the DHCP server sends a reply acknowledging this has been completed, and the device can start using the IP Address (**DHCPACK**).



QUIES

What is the technical term for dividing a network up into smaller pieces?

subnetting

How many **bits** are in a subnet mask?

32

What is the range of a section (octet) of a subnet mask?

0-255

What address is used to identify the start of a network?

network address

What address is used to identify devices within a network?

host address

What does ARP stand for?

address resolution protocol

What category of ARP Packet asks a device whether or not it has a specific IP address?

request

What address is used as a physical identifier for a device on a network?

mac address

What address is used as a logical identifier for a device on a network?

ip address

What type of DHCP packet is used by a device to **retrieve an IP address**?

dhcp discover

What type of DHCP packet does a device **send once it has been offered an IP address** by the DHCP server?

dhcp request

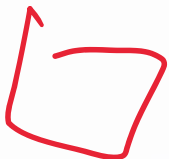
Finally, **what is the last** DHCP packet that is sent to a device from a DHCP server?

dhcp ack

255.255.255.0
- 172.20.10.10 ✓

172.20.10.0

172.20.10.1, 2, 3, 4
- - - 254 ✓



172.20.10.255

172.20.10.0 network
addr

172.20.10.1-254 - host
addr

172.20.10.255-
gateway

